

Detection and Simple Tracking Method for ADAS using a mmWave Radar

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Abstract — Advanced Driver Assistance Systems (ADAS) are becoming reality now. Several researchers are attempting to innovate new methods for vehicle collision avoidance, automated parking, self-driven cars, etc. Towards this area, this paper proposes a mechanism for the detection of targets and a simple tracking method for tracking the same. An experiment was carried out using 77 GHz mmWave Radar that was mounted on a car. The Radar was used to track the moving vehicles as well as stationary objects in front of the ego vehicle. The Radar raw data was recorded in real time while the ego vehicle was on the move. The raw data was then processed to obtain Range-Doppler Map (RD Map). Further Range vs Speed map was obtained from RD-map. 2D Constant False Alarm Rate (CFAR) algorithm was performed on Range vs Speed map for detection of targets on each frame. Further, CFAR detection map of each processed frame was cumulatively added to form a single Detection and Tracking Map (DTM). DTM shows the detection of new targets as well as the history of older targets as tracks over a period. Older targets are gradually deleted from the DTM by applying the deletion rule. The experiment was conducted with the ego vehicle on the road, moving with a speed of around 30 kmph. The Radar raw data obtained with this setup and the proposed algorithms can be used as an input for development of advanced algorithms for collision avoidance, automated car parking and even for self-driven cars.

Keywords— *FMCW, mmWave Radar, Range Doppler Map, Range-Speed Map, CFAR, Detection Map, Detection and Tracking Map.*

I. INTRODUCTION

In recent years, millimeter wave (mmWave) Radar is a popular choice for the development of various applications in automobiles. These mmWave Radars are generally Frequency Modulated Continuous Wave (FMCW) systems. These Radars are compact, low-power, and available at low cost. These Radars are based on System on Chip (SoC) modules. Most of these Radar modules are designed for a wide temperature range which makes them suitable for different geographical areas and different environmental conditions. The main objective of these Radars are target detection, tracking, classification, finding direction of arrival, etc. Researchers are also using cameras, and ultrasonic sensors in addition to mmWave Radar for better performance. These heterogeneous sensors are used to handle complex traffic scenarios like dense traffic situations, the sudden arrival of intruders, etc. The decision logic system uses Radar output to arrive at right decision. Based on the decision logic system output, vehicle cruise control system controls the vehicle to

maintaining a safe distance from other vehicles/obstacles. The actions can be applying brakes, lane changing, controlling the steering in safe directions, etc.

A detail review on Automotive Radars can be found in [1]. It also provides information on types of automotive Radars, technologies, applications, Radar signal processing techniques, Multi-Input Multi Output (MIMO) Radars in automotive industries, waveform orthogonality techniques, etc. MIMO Radar antenna array, state of the art MMIC semiconductor technologies, signal modulation, signal processing techniques, interferences, compressed sensing, Radar networks, and Machine Learning in Automotive applications are discussed in [2]. An experimental setup for detection of on the road moving vehicles by mounting mmWave Radar on top of the vehicle can be found in [3]. Authors proposed CFAR detection and Micro Doppler Classifier (CDMC) techniques for extraction of features from temporal and spatial information.

Collision avoidance is an important application of mmWave Radar. A collision avoidance system for bicycles and runners has been proposed in [4]. Here detection system is an FMCW mono-pulse radar operating in K band along with a camera. Range-angle measurements from Radar and images from a camera are used to obtain a complete outside scenario. Intruder detection plays an important role in collision avoidance. Intruder detection system using 77 GHz Radar and a camera has been proposed in [5]. The data from the Radar and Camera are used for target detection. Extended Kalman filter, density-based DB Scan clustering algorithm and global nearest neighbour algorithm are used to estimate the new target positions for tracking. Tracking of pedestrian by autonomous vehicle to avoid collision is a challenging task. A solution based on multi sensor data fusion for online pedestrian tracking problem can be found in [6].

A simple heterogeneous sensor consisting of Radar and ultrasonic sensors for collision avoidance can be found in [7]. Based on the measured parameters like distance, the vehicle is controlled/steered to avoid collision. Camera and LiDAR data fusion for collision avoidance has been proposed in [8]. Here the sensor fusion is performed to integrate the detected points from source images using deep learning techniques.

Target detection is said to be automatic if the presence or absence of a target is done by the Radar without manual intervention [9]. Target detection is statistical in nature. In simple terms, the probability that a real target can be detected

in a given environment (signal plus interference and a defined threshold) is called Probability of detection (P_d) [9],[10]. On the other hand, the probability that a false target (due to interference) is detected and declared as a true target is called probability of false alarm (P_{fa}). The False Alarm Time - T_{fa} is the mean time between the interference signal crossing the threshold [9].

In order maintain constant false alarm rate, the threshold needs to be calculated adaptively based on the estimate of the background interference from the radar resolution cells around the test cell which are known as reference cells with the assumption that reference cells do not contain the target. Due to this, the CFAR becomes suboptimal as the estimate of the threshold itself becomes a random variable resulting in CFAR loss. There are many CFAR variants [11]. However, two basic elements are (i) sliding window to estimate the signal characteristics with the Cell under test at the center. (ii) determination of the detection threshold which has to be varied in order to maintain the P_{fa} . Maintaining the P_{fa} is crucial as it is linked P_d . The optimal method is based on the Neyman-Pearson criterion [12],[13].

Classification of targets is also essential and being attempted by many researchers. Use of Artificial Intelligence and Machine Learning are being attempted for target classification. Support Vector Machine (SVM) based techniques for Human-Vehicle classification can be found in [14]. Use of Reinforcement learning for Self-driven Cars can be found in [15]. Classification of Automotive Radar targets using inverse synthetic aperture images can be found in [16]. Direction of Arrival (DoA) of signal from the target is also an important parameter for decision making logic. In this context, a deep learning-based approach for estimation of DoA has been proposed in [17]. Experiment for detection of targets on the road using mmWave Radar can be found in [18].

Based on the above literature survey, it is clear that target detection and tracking using mmWave Radar is simple, cost effective and can be used even in adverse environmental conditions[1],[2]. In [3] authors provided similar setup for data collection under different scenario for detection and classification. In addition to target detection and classification, target tracking is also required. Ultrasound or LiDAR based solution [7],[8] are expensive. Hence the motivation is to collect Radar target data using simple setup using mmWave Radar in real-time at different scenarios. This large set of data is essential for development of algorithms and techniques for collision avoidance, automated vehicle parking and finally self-driven cars. Further, it is very important to develop and implement Radar signal processing algorithm for target detection and tracking. A simple detection and tracking algorithm is presented in this experiment. Here a mmWave Radar module was mounted on the bonnet of the car. The Car was driven on the road and real-time raw Radar data was recorded in a laptop. Radar signal processing algorithms were developed in Matlab for processing the raw data.

Following are the contributions of this paper.

- a. An experiment was conducted on the road with a mmWave 77GHz Radar sensor AWR 1843 BOOST along with Data acquisition Card DCA 1000, both from Texas Instruments. These

modules are mounted on the bonnet of the car for detection and tracking of vehicles/objects in front of it.

- b. Raw I and Q data for different scenarios were recorded in a laptop that was connected to the Radar module.
- c. Range-Doppler map was obtained by taking two-dimensional Fast Fourier Transforms (FFT) on each axis i.e. Range axis and Doppler axis. The resultant map is Range-Doppler Map (RD-Map) Later Range-Speed map was obtained from RD Map. Further 2D CFAR was performed on the Range-Speed map to obtain a Detection Map. In the next step, the Detection Map of each frame was cumulatively added to obtain DTM. Vintage detections are deleted from DTM using deletion rule.

This paper is organized as follows:

- a. Section II : Problem Statement
- b. Section III: The experimental setup and Radar Signal processing
- c. Section IV : Analysis of the result
- d. Section V : Concluding remarks

II. PROBLEM STATEMENT

Target detection and tracking is the basic function for any collision avoidance system. The mmWave Radar is well suited for this purpose due to its small size, low cost, and low power operation.

Generally, mmWave Radars are mounted on the front and rear side, as well as both sides of the vehicle. Front-mounted mmWave Radars are used for tracking vehicles/obstacles/stationary objects which are ahead of the ego vehicle. Side-mounted Radars provide information on lateral vehicles/objects and rear-side Radars provide information on approaching vehicles. Collective information provided by these radars are used for collision avoidance.

The primary objectives of this experiment are given below.

1. Data collection: Collection of raw Radar data for different traffic scenarios using mmWave Radar mounted on the bonnet of a car.
2. Target Detection: Targets in the field of view are to be detected by the Radar. The targets can be moving or stationary.
3. Relative speed estimation: Relative speed between ego vehicle and targets to be estimated based on the Doppler response.
4. Target Tracking: Detected targets are to be tracked using simple tracking algorithm. In addition to targets, many false detections can be found in the CFAR detections. False detections are to be deleted and real targets need to be tracked.

III. EXPERIMENTAL SETUP AND RADAR SIGNAL PROCESSING

A. Radar Hardware Setup

Radar hardware setup consist of one each of Texas Instrument AWR1843BOOST (Radar module) and DCA1000 EVM (Data capture card), and a laptop. AWR 1843BOOST and DCA1000 modules are mounted on the bonnet of the car. Radar Module and Data capture card are connected to laptop through USB and ethernet interface. The mmWave Radar setup on test bench is shown in Fig. 1. Radar setup mounted on a car is shown in Fig. 2. Raw data (I and Q) are captured in a laptop while the ego vehicle was moving at a speed of around 30kmph. The captured Raw data is processed in the Radar signal processing module.

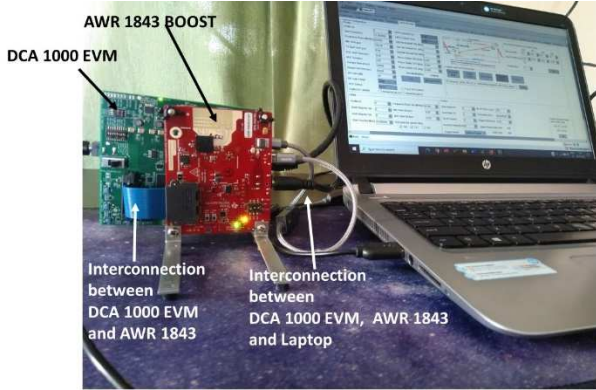


Fig.1. mmWave Radar setup on Test Bench



Fig. 2. mmWave Radar setup on bonnet of the Car

B. AWR 1843BOOST - 77GHz mmWave Radar

The AWR1843BOOST is a mmWave Radar operating in the frequency range of 76-81 GHz for automotive applications. It consists of three transmitters and four receivers. It is an FMCW Radar sensor from Texas Instruments. The Radar module provides a Low Voltage Differential Signaling (LVDS) interface to Data Capture card through high density 60 pin connector. Through this interface, Radar module sends raw I and Q data post Analog to Digital (ADC) conversion to Data capture Card

C. DCA1000 EVM

DCA 1000EVM card from Texas Instruments is used for converting LVDS signals from AWR 1843BOOST module to Gbit ethernet data. For this purpose, AWR 1843BOOST module is mounted on DCA 1000 module. The ethernet port of the DCA 1000 module is connected to a laptop for streaming raw I and Q data to the laptop. The raw I and Q data is in binary form (complex data type) and the Radar signal processing module uses this data for further processing.

D. Radar Signal Processing

Raw data from Radar are stored in the laptop. Raw I and Q Data is divided into several frames. Each frame is an $M \times N$ matrix. Where M is the number of ranges gates per pulse and there are N number of pulses. In this experiment, each frame consists of 256 Ranges gates and 128 pulses. At each instant of the experiment, 1000 frames are captured. Block diagram of the Radar signal processing module is shown in Fig. 3.

The Radar signal Processing module consists of three stages. In the first stage, the raw I and Q data (Fig. 5) is processed. 2D FFT is performed on each frame (256×128). The output of this stage is RD Map (Fig. 6). In the next stage, the RD Map is converted into Range-Speed Map (Fig. 7). Hence the output of this stage indicates relative speed between the ego vehicle and the target vehicle. Based on the speed measurement, the algorithm can decide whether the target vehicle is moving away or approaching. Table-1 provides the relative speed and the inference drawn.

In the second stage, 2D CFAR is performed on the Range-Speed Map. CA CFAR is used with probability of false alarm (P_{fa}) kept at 10^{-4} . The output of 2D FFT i.e., Detection Map and is shown in Fig. 8.

The next stage is Tracking and Deletion Process. Post detection stage, Detection and Tracking Map (DTM) is created. DTM is updated after every frame processing. The algorithm is explained in Fig. 4 for one frame. This process to be repeated for all frames. One such result at K^{th} stage is shown in Fig. 9. To conclude full algorithm is explained in the next section.

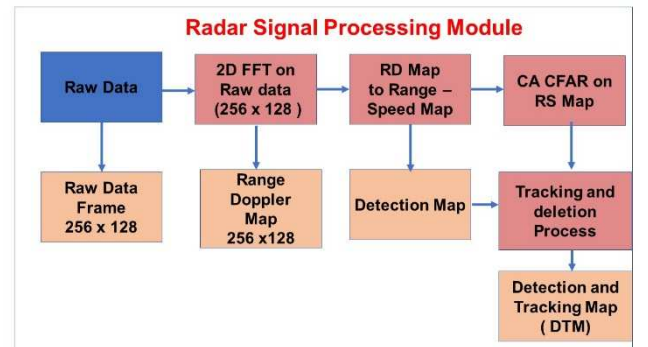


Fig. 3. Block diagram of Radar Signal Processing

TABLE 1. RELATIVE SPEED AND INFERENCE

Relative Speed	Inference
Positive	The target vehicle is moving away from the ego vehicle
Negative	1. Ego Vehicle approaching the target vehicle. In this case the speed of the ego vehicle is higher than the target vehicle. 2. Target vehicle is approaching the ego vehicle in the opposite direction. 3. Ego vehicle is approaching a stationary target.
Zero	Both ego vehicle and target vehicle are moving at the same speed.

E. Radar Target Detection and Tracking Algorithm

Collect Raw IQ data from the Radar hardware

1. Step 1: Select First Frame from Raw I and Q data
2. Step 2: Perform 2D FFT on the frame to obtain Range and Doppler Map (RD-Map)
3. Step 3: Convert Rand Doppler map to Range-Speed map using Doppler information
4. Step 4: Apply 2D CA-CFAR algorithm on Range-Speed to obtain Detection Map (n^{th})
5. Step 5: Update ($n-1$)th DTM with n^{th} Detection Map
6. The Updated ($n-1$)th DTM is now n^{th} DTM
7. If it is the last frame in Raw Radar data then exit, else select the next frame and go to Step 2

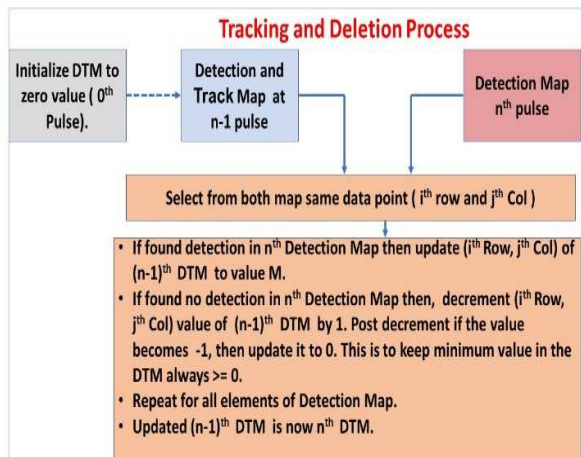


Fig. 4. Tracking and Deletion Process

IV. RESULTS AND DISCUSSIONS

Fig. 5 shows the raw data of a frame. This is the data captured from the Radar modules and plotted using Matlab. Performed 2D FFT on the Raw data to get Range Doppler

Map Fig. 6. RD Map is further processed to get Range-Speed Map Fig. 7. 2D CFAR is performed on Range-Speed map Fig. 8. Fig. 9 shows DTM map result at the K^{th} iteration. One of the targets (Target #1) has a positive relative speed. Hence the target vehicle is moving away from the ego vehicle. On the other hand, the target vehicle (Target #2) has a negative relative speed. Hence this target is approaching the ego vehicle (or the ego vehicle is approaching the target). In this case, the target can be a moving vehicle with speed less than the ego vehicle or target may be approaching the ego vehicle in opposite direction or the detected target may be a stationary object. The DTM provides sufficient information about the recent sustained tracks. In this algorithm, all false tracks will fade away gradually. The rate of fading depends upon the value M . If the value of M is large (say greater than 10) then fading is slow. If the value of M is small (say less than 2), fading is fast. The typical value is between 3-5. The large value of M also means the history of real or false target is maintained for a long time. Since the relative speed of vehicles is small and targets are low maneuvering, this simple solution is sufficient for automotive applications. Further, this DTM can be further processed using AI & ML algorithms for collision avoidance, cruise control, and other applications for self-driven cars.

V. CONCLUSION AND FUTURE WORK

In this paper, a cost-effective setup for collection of real-time data and a simple algorithm for the detection and tracking of automobiles/static objects using mmWave Radar is presented. The radar is built with Texas Instrument AWR1843BOOST and DCA 1000 modules. Experiments are conducted on the road and results are processed to obtain DTM. DTM provides sufficient information about the real target tracks. At the same time, the track's history/false detections fade away slowly.

On the road real-time data that has been collected in this experiment can be used for future research work. Future work is towards analyzing the DTM and develop robust decision logic. AI and ML is proposed for development of decision logic. The output of the decision logic can be used for cruise control. Cruise control system can take necessary actions to avoid a collision.

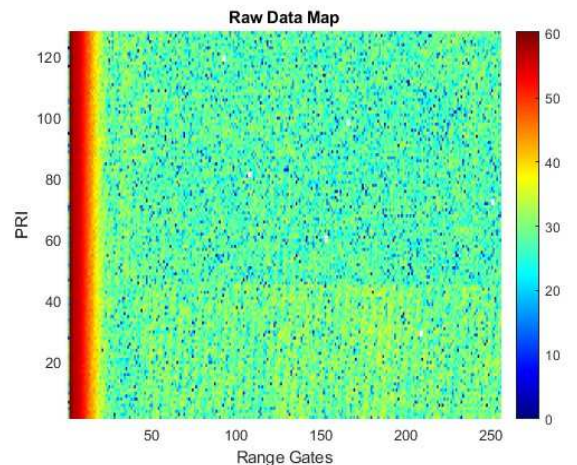


Fig. 5. Raw I and Q Data from Radar Module

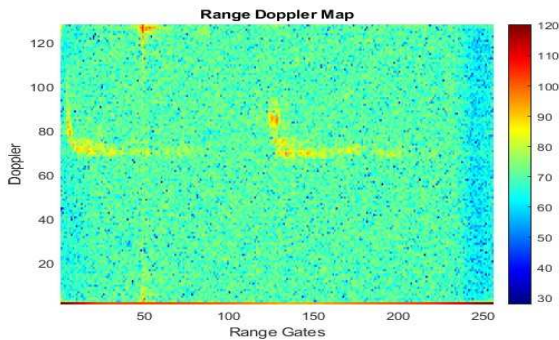


Fig. 6. Range and Doppler Map (128x256)

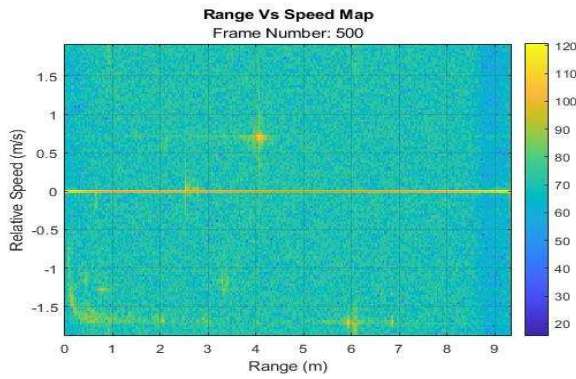


Fig. 7. Range vs Speed Map

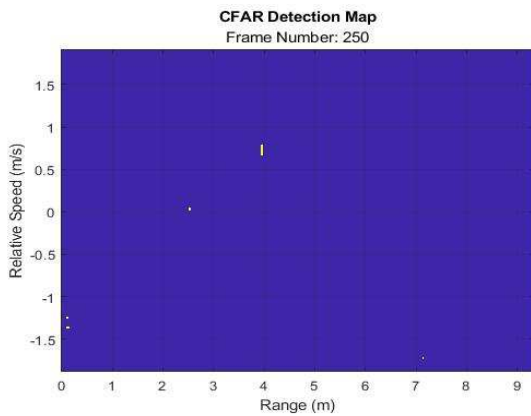


Fig. 8. CFAR Detection Map

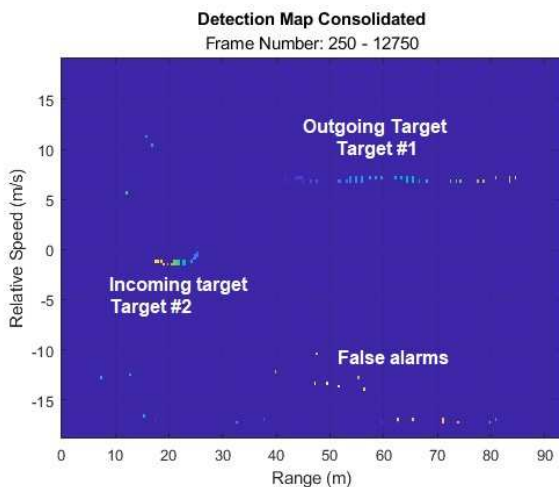


Fig. 9. DTM map Consolidated K^{th} iteration

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